

A DETAILED STUDY OF LEXICAL RESOURCES FOR NLP

# WordNet and Its Variants

Structure • Graph Models • Semantic Similarity • Word Sense Disambiguation  
SentiWordNet • Hindi WordNet • SenticNet • ConceptNet

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Natural Language Processing — Lexical Semantics Module



# What We Will Cover

1

## Foundations

- Lexical resources in NLP
- WordNet: origin & design
- Synsets & semantic relations
- WordNet statistics

2

## Graphs & Similarity

- WordNet as a graph
- Worked numerical examples
- Path, Wu-Palmer, Leacock-Chodorow
- Resnik, Lin, Jiang-Conrad (IC-based)

3

## Disambiguation & Use

- Word Sense Disambiguation
- Lesk algorithm (numerical)
- Modern NLP applications
- Limitations

4

## Beyond WordNet

- SentiWordNet (with scoring)
- Hindi WordNet / IndoWordNet
- SenticNet
- ConceptNet & comparison

# Lexical Resources in NLP

- A lexical resource is a structured repository of words, their meanings, and the relationships between them, built to support computational language processing.
- They supply what raw text alone cannot: sense inventories, semantic relations, sentiment polarity, and world/commonsense knowledge.
- **Broad categories:**
  - Lexical-semantic networks — WordNet, EuroWordNet, IndoWordNet
  - Sentiment/affect lexicons — SentiWordNet, SenticNet, VADER
  - Commonsense knowledge graphs — ConceptNet, ATOMIC
  - Frame & role resources — FrameNet, PropBank, VerbNet
- They act as a bridge between symbolic linguistic knowledge and statistical / neural NLP pipelines.

## Why they still matter in the LLM era

- Ground neural models with explicit, auditable semantic structure
- Provide gold-standard evaluation sets for similarity & relatedness
- Enable low-resource and multilingual transfer (e.g. IndoWordNet)
- Supply interpretable features for explainable AI systems
- Support symbolic + neural hybrid architectures (knowledge-enhanced LLMs)

# What is WordNet?

- WordNet is a large lexical database of English developed at Princeton University, organizing nouns, verbs, adjectives, and adverbs into sets of cognitive synonyms called synsets.
- Each synset expresses a distinct concept and is linked to other synsets through conceptual-semantic and lexical relations.
- It functions simultaneously as a dictionary (glosses/definitions) and a thesaurus (synonym sets), but its network structure makes it far more useful than either alone.
- **Result: a combined structure of a graph and a lexicon — ideal for computational semantics.**

## Quick Facts

|                 |                                        |
|-----------------|----------------------------------------|
| Creator         | George A. Miller & team                |
| Institution     | Princeton University                   |
| Started         | 1985                                   |
| License         | Free, open for research/commercial use |
| Current version | WordNet 3.1                            |
| Format          | Lexicographer files + compiled DB      |

# Synsets: The Atomic Unit of WordNet

- A synset (synonym set) groups words that are interchangeable in some context, representing a single, distinct sense.
- **Each synset contains:**
  - A set of synonymous lemmas (word forms)
  - A gloss — a short definition
  - Usually one or more example sentences
  - A unique synset offset / sense ID
- A word can belong to multiple synsets — this is exactly how WordNet encodes polysemy (multiple senses of one word).

## Example Synset — “Car” (noun, sense 1)

|                 |                                                                                               |
|-----------------|-----------------------------------------------------------------------------------------------|
| <b>Synonyms</b> | car, auto, automobile, machine, motorcar                                                      |
| <b>POS</b>      | Noun                                                                                          |
| <b>Gloss</b>    | <i>“a motor vehicle with four wheels; usually propelled by an internal combustion engine”</i> |
| <b>Example</b>  | <i>“he needs a car to get to work”</i>                                                        |
| <b>Hypernym</b> | motor vehicle                                                                                 |
| <b>Hyponyms</b> | convertible, sedan, hatchback, cab ...                                                        |

# Semantic Relations Encoded in WordNet

| Relation   | Direction                           | Applies to      | Example        |
|------------|-------------------------------------|-----------------|----------------|
| Hypernymy  | General ← Specific (IS-A up)        | Nouns, Verbs    | vehicle ← car  |
| Hyponymy   | General → Specific (IS-A down)      | Nouns, Verbs    | car → sedan    |
| Meronymy   | Whole → Part (HAS-A)                | Nouns           | car → engine   |
| Holonymy   | Part → Whole                        | Nouns           | engine → car   |
| Antonymy   | Opposite meaning                    | Adj, Adv, Nouns | hot ↔ cold     |
| Troponymy  | Manner IS-A (specific way of doing) | Verbs           | whisper → talk |
| Entailment | Verb A implies Verb B               | Verbs           | snore → sleep  |

*Nouns and verbs are organized into separate hypernym/hyponym hierarchies (IS-A trees); adjectives and adverbs are organised around antonymy and “satellite” clusters instead of a tree.*

# WordNet 3.1 — Database Statistics

| POS          | Unique Strings | Synsets  | Total Word-Sense Pairs |
|--------------|----------------|----------|------------------------|
| Noun         | 1,17,798       | 82,115   | 1,46,312               |
| Verb         | 11,529         | 13,767   | 25,047                 |
| Adjective    | 21,479         | 18,156   | 30,002                 |
| Adverb       | 4,481          | 3,621    | 5,580                  |
| <b>Total</b> | 1,55,287       | 1,17,659 | 2,06,941               |

## Reading the Numbers

- Average polysemy (senses/word) for nouns  $\approx 1.24$ ; for the most-used vocabulary it is far higher.
- Nouns dominate the database — reflecting the taxonomic, IS-A driven design.
- $\approx 2,07,000$  word-sense pairs give WordNet its fine-grained sense inventory used in WSD benchmarks (e.g. SemEval, SenseEval).

# WordNet as a Graph — Formal View

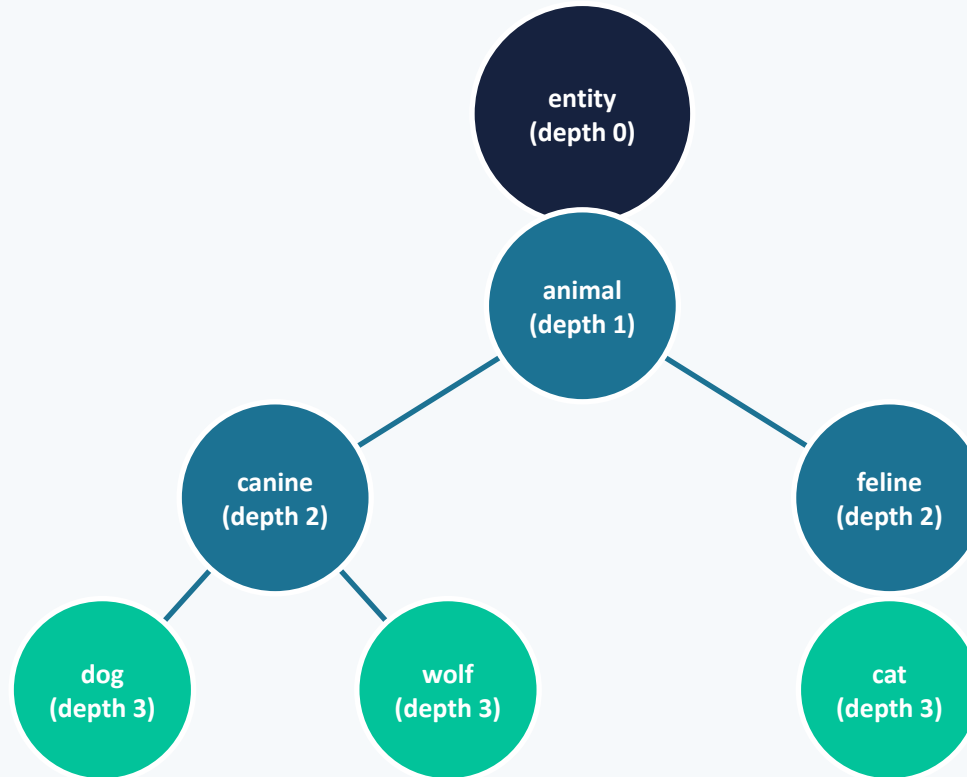
- **WordNet can be formally modelled as a graph  $G = (V, E)$ :**
  - $V$  (vertices) = synsets (concepts)
  - $E$  (edges) = semantic relations (hypernymy, meronymy, antonymy ...), each typed and often directed
- The noun and verb hierarchies form (almost) a Directed Acyclic Graph (DAG) rooted at an abstract node such as “entity”.
- **Key graph properties used computationally:**
  - Depth of a node — distance from the root
  - Path length between two nodes — shortest number of edges
  - Lowest Common Subsumer (LCS) — deepest shared ancestor of two nodes
- These graph properties are precisely what power semantic-similarity algorithms (next section).

## Why “graph”, not just “hierarchy”?

- A synset can have more than one hypernym (multiple inheritance) — so it is a DAG, not a strict tree.
- Multiple relation types create a multi-relational / heterogeneous graph.
- This enables graph algorithms: shortest path, random walks, PageRank-style ranking, spreading activation, and Graph Neural Network embeddings of concepts.



# Worked Example — A Small Taxonomy Fragment



## Reading this fragment

- This is the hypernym chain PrincetonWordNet actually uses for canines and felines (simplified).
- Edges = hypernym / hyponym relation.
- “dog” and “wolf” share the parent “canine” — their Lowest Common Subsumer (LCS).
- “dog” and “cat” share only “animal” as their LCS — a longer path, hence lower similarity.
- **We use this exact fragment for the similarity numericals on the next slides.**

# Depths, Paths & Lowest Common Subsumers

| Synset | Depth from root |
|--------|-----------------|
| entity | 0               |
| animal | 1               |
| canine | 2               |
| feline | 2               |
| dog    | 3               |
| wolf   | 3               |
| cat    | 3               |

| Pair         | Shortest Path (edges)             | LCS    | LCS Depth |
|--------------|-----------------------------------|--------|-----------|
| dog – wolf   | 2 (dog→canine→wolf)               | canine | 2         |
| dog – cat    | 4 (dog→canine→animal→feline→cat)  | animal | 1         |
| wolf – cat   | 4 (wolf→canine→animal→feline→cat) | animal | 1         |
| dog – animal | 2 (dog→canine→animal)             | animal | 1         |

Maximum taxonomy depth here  $D = 3$  (used later in the Leacock-Chodorow formula). In the full Princeton WordNet noun hierarchy, the maximum depth is 19–20.

*These depth and path values feed directly into every similarity metric on the following slides.*

# Measuring Similarity Between Words Using WordNet

- Because synsets sit inside a structured graph, we can turn “how similar are two word senses” into a measurable graph / information-theoretic quantity.

## Edge-Counting (Path-based)

*Use the graph structure only — shortest path length and depth.*

- Path Similarity
- Wu-Palmer Similarity (WUP)
- Leacock-Chodorow Similarity (LCH)

## Information Content (IC-based)

*Combine the graph with corpus statistics — how “informative” a concept is.*

- Resnik Similarity
- Lin Similarity
- Jiang-Conrath Similarity (distance)

## Gloss / Feature-based

*Compare textual glosses or feature overlap rather than pure structure.*

- Lesk-based similarity
- Extended gloss overlap (Banerjee & Pedersen)

# Path Similarity & Wu-Palmer Similarity — Formulas

## 1. Path Similarity

$$\text{simpath}(c1, c2) = 1 / (\text{pathlen}(c1, c2) + 1)$$

- $\text{pathlen}(c1, c2)$  = number of edges on the shortest path connecting synsets  $c1$  and  $c2$  in the IS-A graph.
- Range: (0, 1]. A word compared with itself gives  $\text{pathlen} = 0 \rightarrow \text{similarity} = 1$ .
- Simple and fast, but ignores how deep in the hierarchy the path lies — two edges near the root feel “closer” than two edges near the leaves, yet Path Similarity treats them identically.

## 2. Wu-Palmer Similarity (WUP)

$$\text{simwup}(c1, c2) = 2 \times \text{depth}(\text{LCS}) / (\text{depth}(c1) + \text{depth}(c2))$$

- LCS = Lowest Common Subsumer of  $c1$  and  $c2$ ; depth is measured from the taxonomy root.
- Weighs similarity by how deep the shared ancestor is — a deep LCS (very specific shared concept) gives higher similarity than a shallow one.
- Range: (0, 1]. Widely used as it correlates well with human judgment and is cheap to compute.

# Computing Path & Wu-Palmer Similarity

Using the taxonomy fragment from Slide 9 (depths: dog = wolf = cat = 3, canine = feline = 1, animal = 1, entity = 0):

## dog – wolf

Shortest path length = 2 edges

LCS = "canine" (depth = 2)

**Path Similarity** =  $1 / (2 + 1) = 0.333$

**Wu-Palmer** =  $2 \times 2 / (3 + 3) = 0.667$

## dog – cat

Shortest path length = 4 edges

LCS = "animal" (depth = 1)

**Path Similarity** =  $1 / (4 + 1) = 0.200$

**Wu-Palmer** =  $2 \times 1 / (3 + 3) = 0.333$

Interpretation: dog–wolf (0.333 / 0.667) scores noticeably higher than dog–cat (0.200 / 0.333) on both metrics — matching our intuition that dogs and wolves are more closely related.

# Leacock-Chodorow Similarity (LCH)

## Formula

$$\text{simLCH}(c1, c2) = -\log(\text{pathlen}(c1, c2) / (2 \times D))$$

- $D$  = maximum depth of the taxonomy (a constant for the whole hierarchy).
- Scales the raw path length against the overall size of the hierarchy, so it is comparable across different branches of WordNet.
- Higher score = more similar; the score is unbounded above (as  $\text{pathlen} \rightarrow 0$ ).

## Numerical Example ( $D = 3$ , from Slide 10)

dog – wolf ( $\text{pathlen} = 2$ )

$$\text{simLCH} = -\log(2/6) = -\log(0.333) = 1.099$$

dog – cat ( $\text{pathlen} = 4$ )

$$\text{simLCH} = -\log(4/6) = -\log(0.667) = 0.405$$

*Note: natural log used here; some implementations use  $\log_{10}$  — always state the base when reporting LCH scores.*

# Resnik, Lin & Jiang-Conrath Similarity

*Information Content:  $IC(c) = -\log P(c)$ , where  $P(c)$  is the probability of encountering concept  $c$  (or its descendants) in a large corpus. Rarer / more specific concepts  $\rightarrow$  higher  $IC$ .*

## Resnik Similarity

$$\text{simres}(c1, c2) = IC(\text{LCS}(c1, c2))$$

Similarity is simply the information content of the two concepts' lowest common subsumer. Ignores individual  $IC$  of  $c1, c2$ .

## Lin Similarity

$$\text{simLin}(c1, c2) = \frac{2 \times IC(\text{LCS})}{IC(c1) + IC(c2)}$$

Normalizes Resnik's score by the concepts' own information content — range (0,1], widely used.

## Jiang-Conrath (distance)

$$\begin{aligned} \text{distJC}(c1, c2) &= IC(c1) + IC(c2) - 2 \times IC(\text{LCS}) \\ \text{simJC} &= 1 / \text{distJC} \end{aligned}$$

A distance measure by design; inverted to a similarity. Very sensitive to corpus statistics.

# Computing Resnik, Lin & Jiang-Conrath Similarity

Assume the following illustrative Information Content values (derived from a hypothetical corpus, natural log):

| Concept | IC(c) |
|---------|-------|
| dog     | 6.90  |
| cat     | 7.10  |
| wolf    | 7.80  |
| canine  | 3.20  |
| animal  | 1.90  |

| Pair     | LCS    | Resnik | Lin   | JC dist. | simJC |
|----------|--------|--------|-------|----------|-------|
| dog-cat  | animal | 1.90   | 0.271 | 10.20    | 0.098 |
| dog-wolf | canine | 3.20   | 0.435 | 8.30     | 0.120 |

## Worked calculation — dog vs cat

$$\text{Resnik} = \text{IC}(\text{animal}) = 1.90$$

$$\text{Lin} = 2 \times 1.90 / (6.90 + 7.10) = 3.80 / 14.00 = 0.271$$

$$\text{JC distance} = 6.90 + 7.10 - 2 \times 1.90 = 10.20 \rightarrow \text{simJC} = 1 / 10.20 = 0.098$$

Values are illustrative (assumed IC), used purely to demonstrate the arithmetic of each formula.



# Similarity Measures at a Glance

| Measure          | Type                | Needs Corpus Stats? | Sensitive to Depth? | Typical Use                            |
|------------------|---------------------|---------------------|---------------------|----------------------------------------|
| Path Similarity  | Edge counting       | No                  | No                  | Fast, coarse similarity                |
| Wu-Palmer (WUP)  | Edge counting       | No                  | Yes                 | General-purpose, popular default       |
| Leacock-Chodorow | Edge counting       | No                  | Yes (via D)         | Cross-branch comparability             |
| Resnik           | Information Content | Yes                 | Indirectly          | Concept specificity ranking            |
| Lin              | Information Content | Yes                 | Indirectly          | Normalized, high correlation w/ humans |
| Jiang-Conrath    | Information Content | Yes                 | Indirectly          | Fine-grained distance-based tasks      |

# The WSD Problem

- Word Sense Disambiguation (WSD) is the task of identifying which sense of a polysemous word is being used in a given context.
- Example: “bank” can mean a financial institution or the land beside a river.
- **WordNet is the standard sense inventory for WSD — each candidate “sense” is literally a WordNet synset with its own gloss.**
- **Main families of WSD approaches:**
  - Knowledge-based — Lesk algorithm, graph-based (WordNet structure)
  - Supervised — classifiers trained on sense-tagged corpora (SemCor)
  - Unsupervised / embedding-based — contextual embeddings (BERT) clustered against WordNet glosses

## Example — Polysemy of “bank” in WordNet

**bank#n#1**

*a financial institution that accepts deposits and channels money into lending*

**bank#n#2**

*sloping land beside a body of water*

**bank#n#9**

*a supply held in reserve (e.g. blood bank)*

# The Lesk Algorithm

## Original Lesk (1986)

- Compares the gloss of each sense of the target word against the glosses of every sense of every neighbouring content word.
- Chooses the sense-combination with maximum total word overlap across all glosses.
- Very accurate on short definitions but computationally expensive — combinatorial explosion across neighbours.

## Simplified Lesk

- Compares the gloss (and examples) of each candidate sense of the target word directly against the words in the surrounding context sentence.
- Picks the sense whose gloss has the highest word-overlap with the context.
- Much faster; the de-facto version used in most NLP toolkits (e.g. NLTK).

## Core scoring idea (both variants)

$\text{score}(\text{sense}) = | \text{words}(\text{gloss}(\text{sense})) \cap \text{words}(\text{context}) | \rightarrow \text{choose sense} = \text{argmax}(\text{score})$

# Simplified Lesk — Worked Example

Sentence: "I went to the bank to deposit cash." Target word: bank

Context words (target & stopwords removed): { went, deposit, cash }

bank#n#1

SELECTED

*Gloss: "financial institution that accepts deposits and channels money into lending activities"*

Overlap with context: deposit(s)

**Score = 1**

bank#n#2

*Gloss: "sloping land beside a body of water"*

Overlap with context: (none)

**Score = 0**

Result:  $\text{argmax}(\text{score}) \rightarrow \text{bank\#n\#1}$  (financial institution) is selected as the correct sense, since "deposit" overlaps with its gloss while sense 2's gloss shares no context words.

# Graph-Based WSD Using WordNet

- Instead of comparing glosses only, graph-based WSD exploits the WordNet relation graph directly.
- **Typical pipeline:**
  - Build a subgraph containing all candidate synsets of every content word in the sentence, plus their neighbours (hypernyms, hyponyms, etc.)
  - Run a graph-centrality algorithm — e.g. Personalized PageRank / degree centrality — over this subgraph
  - For each word, select the candidate synset with the highest centrality score
- Intuition: the “correct” combination of senses is the one that is most mutually connected — correct senses reinforce each other through shared relations.

## Key algorithms in this family

- Structural Semantic Interconnections (SSI) — Navigli & Velardi, 2005
- Personalized PageRank WSD — Agirre & Soroa, 2009
- Degree centrality WSD — Navigli & Lapata, 2010
- These generally outperform simple gloss-overlap (Lesk) methods on standard WSD benchmarks, at higher computational cost.

# WordNet's Role in Modern NLP

1

## Query Expansion & IR

Expanding search queries with synonyms/hypernyms to improve recall in search engines.

2

## Knowledge-Enhanced LLMs

Injecting WordNet relations into models like SenseBERT / KnowBERT for sense-aware representations.

3

## Embedding Retrofitting

Post-processing word embeddings (word2vec/GloVe) using WordNet relations to pull synonyms closer (Faruqui et al.).

4

## Text Simplification

Replacing rare/complex words with simpler synonyms drawn from the same synset.

5

## Machine Translation

Interlingual pivot via multilingual WordNets (e.g. Open Multilingual WordNet) for low-resource language pairs.

6

## Ontology & Taxonomy Building

Seeding domain ontologies with WordNet's IS-A backbone before specializing.

7

## Question Answering

Sense disambiguation of question terms to match correct answer passages.

8

## Chatbots & Conversational AI

Synonym/paraphrase generation and intent normalization.

# Limitations of WordNet



Static & manually curated — struggles to keep pace with neologisms, slang, and domain-specific terms



Coarse-grained sense distinctions — many WordNet senses are too fine or too subtle for downstream tasks



English-centric core — other languages rely on smaller, less mature WordNets (though this is improving, e.g. IndoWordNet)



Sparse coverage of named entities, multi-word expressions, and domain jargon



No native sentiment, affect, or commonsense knowledge — motivated SentiWordNet, SenticNet, ConceptNet



Relation set is limited to a fixed taxonomy — cannot represent arbitrary real-world facts (unlike ConceptNet)

# SentiWordNet

- SentiWordNet is a lexical resource built on top of WordNet in which every synset is assigned three sentiment scores.
- **Pos(s), Neg(s), Obj(s) — Positivity, Negativity, and Objectivity, each in  $[0,1]$ , and  $\text{Pos(s)} + \text{Neg(s)} + \text{Obj(s)} = 1$**
- Constructed semi-automatically using a combination of:
  - A seed set of manually labelled synsets
  - A semi-supervised classifier committee trained to propagate scores through WordNet's synonymy/relation graph
  - Random-walk style propagation across the WordNet graph
- Widely used in sentiment analysis, opinion mining, and review classification pipelines.

## Example Entries

| Synset            | Pos   | Neg   | Obj   |
|-------------------|-------|-------|-------|
| good#a#1          | 0.750 | 0.000 | 0.250 |
| bad#a#1           | 0.000 | 0.625 | 0.375 |
| disappointing#a#1 | 0.000 | 0.625 | 0.375 |
| excellent#a#1     | 0.875 | 0.000 | 0.125 |



# Scoring a Sentence with SentiWordNet

Sentence: "The movie was good, but the ending was disappointing."

| Sentiment Word | Synset            | Pos   | Neg   | Obj   |
|----------------|-------------------|-------|-------|-------|
| good           | good#a#1          | 0.750 | 0.000 | 0.250 |
| disappointing  | disappointing#a#1 | 0.000 | 0.625 | 0.375 |

## Aggregate Sentence Score

$$\Sigma \text{ Pos} = 0.750 + 0.000 = 0.750$$

$$\Sigma \text{ Neg} = 0.000 + 0.625 = 0.625$$

$$\text{Score} = \Sigma \text{Pos} - \Sigma \text{Neg}$$

$$= 0.750 - 0.625 = +0.125$$

## Interpretation

- A small positive aggregate score (+0.125) indicates the sentence is mildly positive overall — the praise for the movie slightly outweighs the criticism of the ending.
- Common aggregation strategies: simple sum/average of Pos–Neg per token, or weighting by POS tag / negation handling for more accurate results.
- Real pipelines also apply negation scope detection (e.g. “not good” flips polarity) before aggregating.

# Hindi WordNet / IndoWordNet

- Hindi WordNet was developed by CFILT, IIT Bombay, using the expansion approach — linking Hindi synsets to the corresponding Princeton WordNet (English) synsets rather than building an independent hierarchy from scratch.
- This preserves cross-lingual structural consistency: the same synset ID links equivalent concepts across languages.
- **IndoWordNet extends this to 18+ Indian languages (Hindi, Marathi, Sanskrit, Bengali, Punjabi, Tamil, Telugu, Kannada, etc.), all linked through the Hindi WordNet as the pivot.**
- Contains relations analogous to Princeton WordNet — hypernymy, hyponymy, meronymy — plus additional entries for compounds and idioms specific to Indian languages.

## Snapshot

|                          |                                             |
|--------------------------|---------------------------------------------|
| Developer                | CFILT, IIT Bombay                           |
| Approach                 | Expansion (linked to Princeton WN)          |
| Synsets (Hindi)          | ~40,000+                                    |
| Languages in IndoWordNet | 18+ Indian languages                        |
| Pivot language           | Hindi                                       |
| Use cases                | MT, cross-lingual IR, regional-language NLP |

# SenticNet

- SenticNet is a concept-level sentiment resource that goes beyond word-level polarity to represent multi-word common-sense concepts (e.g. “lose\_friend”, “lose\_relationship”).
- Built via “sentic computing” — combining AI, linguistics, and affective computing, using AffectiveSpace (a vector space) to generalize sentiment from labelled seed concepts to related ones.
- **Each concept carries:**
  - Polarity value — continuous score in  $[-1, +1]$
  - Semantics — related concepts
  - Four “sentic” affective dimensions (Introspection, Temper, Attitude, Sensitivity) inspired by Plutchik's model of emotion
- SenticNet 7/8 covers on the order of ~2,00,000 natural-language concepts, and is widely used in aspect-based sentiment analysis and multimodal sentiment analysis.

## Example Concept Entry

|                  |                       |
|------------------|-----------------------|
| Concept          | “celebrate_occasion”  |
| Polarity         | +0.82                 |
| Introspection    | +0.6                  |
| Temper           | +0.3                  |
| Attitude         | +0.7                  |
| Sensitivity      | +0.4                  |
| Related concepts | party, joy, festivity |

# ConceptNet

- ConceptNet is a large, multilingual commonsense knowledge graph, originally spun out of the MIT Media Lab's Open Mind Common Sense project.
- Unlike WordNet's fixed taxonomic relations, ConceptNet encodes everyday commonsense facts using ~30+ relation types, e.g.:
  - IsA, PartOf, UsedFor, CapableOf, Causes, HasProperty, AtLocation, MotivatedByGoal
- Assertions come from crowdsourcing, games with a purpose, expert resources (including WordNet itself), and other structured sources — then merged into a single graph.
- Each edge carries a confidence/weight score, supporting probabilistic reasoning rather than strict logical entailment.
- Powers the ConceptNet Numberbatch embeddings, widely used for commonsense-aware NLP tasks.

## Snapshot & Example Edges

|                    |                                 |
|--------------------|---------------------------------|
| Nodes (concepts)   | ~8 million (v5.7, multilingual) |
| Edges (assertions) | ~21 million                     |
| Languages          | 300+                            |

(dog, IsA, animal)

(dog, CapableOf, bark)

(bank, UsedFor, store money)

(fire, Causes, burn)

# WordNet & Its Variants — Side by Side

| Resource                    | Core Focus                     | Structure                                  | Scale                                 | Typical Application               |
|-----------------------------|--------------------------------|--------------------------------------------|---------------------------------------|-----------------------------------|
| WordNet                     | Lexical semantics              | Synsets + IS-A / part-whole relations      | ~1,17,000 synsets                     | Similarity, WSD, thesaurus lookup |
| SentiWordNet                | Sentiment polarity             | WordNet synsets + Pos/Neg/Obj scores       | ~1,17,000 scored synsets              | Sentiment / opinion mining        |
| Hindi WordNet / IndoWordNet | Multilingual lexical semantics | Linked synsets across Indian languages     | ~40,000+ Hindi synsets; 18+ languages | Regional-language NLP, MT         |
| SenticNet                   | Concept-level affect           | Concepts + polarity + 4 sentic dimensions  | ~2,00,000 concepts                    | Aspect-based sentiment analysis   |
| ConceptNet                  | Commonsense knowledge          | Multi-relational graph, 30+ relation types | ~8M nodes / ~21M edges                | Commonsense reasoning, embeddings |

# Other Notable Lexical Resources

F

## FrameNet

Organizes meaning around semantic frames (situations) and the roles participants play within them, e.g. the Commerce\_buy frame with Buyer, Seller, Goods.

P

## PropBank

Annotates verbs with numbered semantic argument roles (Arg0, Arg1 ...) for shallow semantic role labelling, widely used in SRL systems.

V

## VerbNet

A verb lexicon organized by Levin verb classes, linking syntactic frames to semantic roles — often used alongside WordNet and PropBank.

B

## BabelNet

A very large multilingual encyclopedic dictionary, merging WordNet with Wikipedia and Wiktionary across 500+ languages.

E

## EuroWordNet

A multilingual database linking WordNets of several European languages through a shared Inter-Lingual Index (ILI).

A

## ATOMIC

A commonsense knowledge graph focused on everyday inferential “if-then” causal and social knowledge, complementary to ConceptNet.

# Key Takeaways

- 1 WordNet models word meaning as a graph of synsets connected by semantic relations — this graph view is what enables computational similarity and WSD.
- 2 Edge-counting measures (Path, WUP, LCH) are fast and structure-only; Information-Content measures (Resnik, Lin, JCN) add corpus-driven nuance.
- 3 The Lesk algorithm and graph-centrality methods turn WordNet's glosses and structure into practical, working WSD systems.
- 4 WordNet's variants — SentiWordNet, Hindi WordNet/IndoWordNet, SenticNet, ConceptNet — extend the same core idea to sentiment, multilinguality, affect, and commonsense knowledge.
- 5 Even in the era of large language models, these resources remain valuable for grounding, evaluation, interpretability, and low-resource / multilingual NLP.

# Select References & Further Reading

- Miller, G. A. (1995). WordNet: A Lexical Database for English. *Communications of the ACM*, 38(11), 39–41.
- Fellbaum, C. (ed.) (1998). *WordNet: An Electronic Lexical Database*. MIT Press.
- Wu, Z. & Palmer, M. (1994). Verb Semantics and Lexical Selection. *ACL*.
- Leacock, C. & Chodorow, M. (1998). Combining Local Context and WordNet Similarity for Word Sense Identification.
- Resnik, P. (1995). Using Information Content to Evaluate Semantic Similarity in a Taxonomy. *IJCAI*.
- Lin, D. (1998). An Information-Theoretic Definition of Similarity. *ICML*.
- Jiang, J. & Conrath, D. (1997). Semantic Similarity Based on Corpus Statistics and Lexical Taxonomy. *ROCLING X*.
- Lesk, M. (1986). Automatic Sense Disambiguation Using Machine Readable Dictionaries. *SIGDOC*.
- Baccianella, S., Esuli, A. & Sebastiani, F. (2010). SentiWordNet 3.0. *LREC*.
- Bhattacharyya, P. (2010). IndoWordNet. *LREC*.
- Cambria, E. et al. (2022). SenticNet 7: A Commonsense-based Neurosymbolic AI Framework for Explainable Sentiment Analysis.
- Speer, R., Chin, J. & Havasi, C. (2017). ConceptNet 5.5: An Open Multilingual Graph of General Knowledge. *AAAI*.